

EVALUATING NUMERICAL EXPRESSIONS WITH INTEGER EXPONENTS – PART 2



LESSON INTRODUCTION

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

LEARNING TARGETS

- I can evaluate a numerical expression using the laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How can you use the laws of exponents to evaluate numerical expressions?
- Does order of operations apply when evaluating numerical expressions with exponents?

LESSON NARRATIVE

Students continue building mathematical fluency in applying the laws of exponents to evaluate numerical expressions with exponents. It is reinforced for students that laws can be applied differently but generate equivalent expressions that have the same value. Students discover order of operations is still applicable when evaluating expressions with exponents.

COMPONENT SUMMARY

4.5.1 WARM-UP (~5 MIN.)

Reflect on growth mindset video

4.5.3 TRY IT (10 MIN.)

Use different strategies to evaluate an expression

4.5.5 YOUR TURN! (5-10 MIN.)

Independent practice evaluating numerical expressions

4.5.2 GUIDED PRACTICE (10 MIN.)

Analyze and synthesize two students' processes used to evaluate expressions

4.5.4 COLLABORATION (15 MIN.)

Evaluate numerical expressions

4.5.6 WRAP-UP (~5 MIN.)

Reflect on understanding of the learning target

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Zero exponent
- Power of a power
- Power of products
- Power of quotients
- Identity exponent
- Negative exponent
- Product of powers
- Quotient of powers

MATERIALS

- Growth Mindset: Mistakes Help You Grow video
- (Optional) Highlighters
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize that the order of operations still applies when simplifying expressions with exponents.

THINGS TO CONSIDER

- This lesson does not require calculators.
- Consider exploring all correct ways the laws of exponents can be used to evaluate each expression.
- Consider allowing students to use highlighters or markers and chart paper to complete 4.5.4 Collaboration.

LITERACY AND LANGUAGE ROUTINES

Number Talk

Consider implementing this routine during 4.5.3 Try It. After showing the problem, give students a few minutes to quietly think and give a signal when they have an answer and a strategy. Select students to share different strategies. Ask, “Who thought about it a different way?” Record explanations for all to see. Prompt students to provide more details about why they decided to approach a problem the way they did.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 4.5.4 Collaboration, students evaluate an expression using a different strategy than their partner. Then they discuss similarities and differences in their work. In doing so, students will recognize the patterns and structure in applying laws of exponents to evaluate numerical expressions.

DIFFERENTIATE INSTRUCTION

4.5.2 Guided Practice provides a visual entry point by having students highlight, circle, or underline each step the sample students took as they simplified the expressions. Provide an auditory entry point by reading the completed paragraph in question 1b aloud.

Additional differentiation opportunities are denoted throughout the lesson guide.

Danny's Work	Reina's Work
$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$	$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$
$= \frac{\left(\frac{3}{4}\right)^{-3+0}}{\left(\frac{3}{4}\right)^{-2}}$	$= \frac{\left(\frac{3}{4}\right)^{-3} \cdot 1}{\left(\frac{3}{4}\right)^{-2}}$
$= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$	$= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$
$= \left(\frac{3}{4}\right)^{-3-(-2)}$	$= \left(\frac{3}{4}\right)^{-3-(-2)}$
$= \left(\frac{3}{4}\right)^{-1}$	$= \left(\frac{3}{4}\right)^{-1}$
$= \frac{4}{3}$	$= \frac{4}{3}$

4.5.1 WARM-UP: MISTAKES HELP YOU GROW

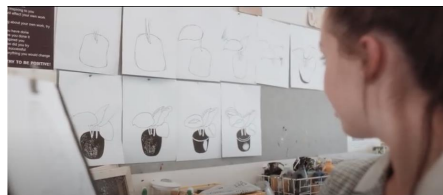
ENRICHMENT

How could you apply this example to your work in math class?
How do you feel when you make a mistake? Why do you feel this way? Why are making mistakes beneficial?



4.5.1 Warm-Up: Mistakes Help You Grow

- Two artists are drawing a plant. As they go through the process, both artists make mistakes and start over on their drawings. One of the artists crumples their papers up and throws them on the floor. The other artist hangs them on the wall next to where she is working.



What do you think is the purpose of hanging up the drawings with mistakes?

Answers will vary. Hanging up drawings with mistakes allows us to see drawings that need edits. It also allows us to remember that mistakes happen, and lead to growth.



4.5.2 Guided Practice: Evaluating Exponential Expressions

1. Danny and Reina evaluated the expression $\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$. They found the same value but used different processes. Their work is shown.

Danny's Work	Reina's Work
$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3+0}}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$ $= \left(\frac{3}{4}\right)^{-3-(-2)}$ $= \left(\frac{3}{4}\right)^{-1}$ $= \frac{4}{3}$	$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3} \cdot 1}{\left(\frac{3}{4}\right)^{-2}}$ $= \frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$ $= \left(\frac{3}{4}\right)^{-3-(-2)}$ $= \left(\frac{3}{4}\right)^{-1}$ $= \frac{4}{3}$

- a. With your partner, discuss and compare how each student evaluated the expression.
Answers will vary. Discussions may include the difference in properties applied in the first step and the similarities in steps after that.

- b. Use words or phrases from the bank to complete the descriptions of each student's work. Some words or phrases will be used more than once and others not at all

zero	identity
power of a power	negative
power of a product	product of powers
power of a quotient	quotient of powers

Danny applied the product of powers exponent law first by rewriting $\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0$ as $\left(\frac{3}{4}\right)^{-3}$. Then, he applied the quotient of powers exponent law, subtracting the exponents -3 and -2 . Finally, he applied the negative exponent law to find the value of the expression.

Reina applied the zero exponent law first by re-writing $\left(\frac{3}{4}\right)^0$ as 1. Then, she multiplied $\left(\frac{3}{4}\right)^{-3}$ by 1. Next, she applied the quotient of powers exponent law by subtracting the exponents -3 and -2 . Finally, she applied the negative exponent law to find the value of the expression.

While the students applied the laws differently, they both generated equivalent expressions in each step to find the value of the expression.

4.5.2 GUIDED PRACTICE: EVALUATING EXPONENTIAL EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Use observational data from prior lessons to dictate how “guided” this 4.5.2 Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

DIFFERENTIATE INSTRUCTION

Provide a visual entry point by having students highlight, circle, or underline each step the sample students took as they simplified the expressions. Provide an auditory entry point by reading the completed paragraph in Question 1b aloud.

4.5.3 TRY IT: EVALUATING EXPRESSIONS

TEACHER MOVES

Number Talk

After showing the problem, give students a few minutes to quietly think and give a signal when they have an answer and a strategy. Select students to share different strategies. Ask, “Who thought about it a different way?” Record explanations for all to see. Prompt students to provide more details about why they decided on their approach.

ENRICHMENT

What do you notice about the order of the calculations?



4.5.3 Try It: Evaluating Expressions

1. Consider the expression $\frac{\left(\left(\frac{1}{3}\right)^{-5} \cdot \left(\frac{1}{3}\right)^2\right)^2}{\left(\frac{1}{3}\right)^{-4}}$.

- a. Evaluate the expression in two different ways using the strategies described in the table. Show your work under each strategy.

Strategy 1: Start by applying the product of powers law of exponents.	Strategy 2: Start by applying the power of a product law of exponents.
$\frac{\left(\frac{1}{3}\right)^{-5} \cdot \left(\frac{1}{3}\right)^2}{\left(\frac{1}{3}\right)^{-4}}$ $\frac{\left(\frac{1}{3}\right)^{-3}}{\left(\frac{1}{3}\right)^{-4}}$ $\left(\frac{1}{3}\right)^{-3-(-4)}$ $\left(\frac{1}{3}\right)^{-3+4}$ $\left(\frac{1}{3}\right)^1$ 3^2 9	$\frac{\left(\frac{1}{3}\right)^{-5} \cdot \left(\frac{1}{3}\right)^2}{\left(\frac{1}{3}\right)^{-4}}$ $\frac{\left(\frac{1}{3}\right)^{-10} \cdot \left(\frac{1}{3}\right)^4}{\left(\frac{1}{3}\right)^{-4}}$ $\frac{\left(\frac{1}{3}\right)^{-6}}{\left(\frac{1}{3}\right)^{-4}}$ $\left(\frac{1}{3}\right)^{-6-(-4)}$ $\left(\frac{1}{3}\right)^{-6+4}$ $\left(\frac{1}{3}\right)^{-2}$ 3^2 9

- b. Explain whether or not the quotient of powers law of exponents could have been applied first.

The quotient of powers law could not have been applied first because the exponent laws should be applied to the numerator first. Order of operations still apply when simplifying exponents.



4.5.4 Collaboration: Choose Your Own Adventure!

1. Consider the expression $\frac{(2^{-1})^3 \cdot (\frac{1}{2})^{-4}}{2 \cdot 2^2}$.

- a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

- | | |
|--|--|
| ☐ Zero Exponent | ☐ Product of Powers |
| ☐ Identity Exponent | ☐ Power of a Product |
| ☐ Negative Exponent | ☐ Quotient of Powers |
| ☐ Power of a Power | ☐ Power of a Quotient |

- b. Evaluate the expression beginning with a different first step than your partner.

Answers will vary.

$$\frac{(2^{-1})^3 \cdot (\frac{1}{2})^{-4}}{2 \cdot 2^2}$$

$$\frac{(2^{-1})^3 \cdot (2)^4}{2 \cdot 2^2}$$

$$\frac{2^{-3} \cdot 2^4}{2 \cdot 2^2}$$

$$\frac{2}{2 \cdot 2^2}$$

$$\frac{2}{2 \cdot 4}$$

$$\frac{2}{8}$$

$$\frac{1}{4}$$

- c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.

2. Consider the expression $\frac{(\frac{2}{3})^{-5} \cdot ((\frac{2}{3})^3)^9}{((\frac{2}{3})^2)^3}$.

- a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

- | | |
|--|--|
| ☐ Zero Exponent | ☐ Product of Powers |
| ☐ Identity Exponent | ☐ Power of a Product |
| ☐ Negative Exponent | ☐ Quotient of Powers |
| ☐ Power of a Power | ☐ Power of a Quotient |

- b. Evaluate the expression beginning with a different first step than your partner.

Answers will vary.

$$\frac{(\frac{2}{3})^{-5} \cdot ((\frac{2}{3})^3)^9}{((\frac{2}{3})^2)^3}$$

$$\frac{(\frac{2}{3})^{-5} \cdot 1}{((\frac{2}{3})^2)^3}$$

$$\frac{(\frac{2}{3})^{-5}}{((\frac{2}{3})^2)^3}$$

$$\frac{(\frac{2}{3})^5}{(\frac{2}{3})^6}$$

$$(\frac{2}{3})^{-1}$$

$$\frac{3}{2}$$

- c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.

4.5.4 COLLABORATION: CHOOSE YOUR OWN ADVENTURE!

DIFFERENTIATE INSTRUCTION

Consider displaying an anchor chart with examples of each law of exponents for students to reference.

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.

TEACHER MOVES

Encourage students to explain why each law could or could not be applied first. Guide students to make the connection to recognize that they must use the order of operations when simplifying expressions.

DIFFERENTIATE INSTRUCTION

Students may benefit from sentence stems such as:

- "Our methods were similar because we both..."
- "Our methods were different, because..."

4.5.5 YOUR TURN!

ENRICHMENT

- Why is the second expression not a correct first step?
- Which first step did you take?
- Why did you select this as your first step?



4.5.5 Your Turn!

1. Otis is evaluating the expression $\frac{3.2^{-4} \cdot 3.2^0}{(3.2^{-3})^2}$.
- a. Circle the expressions that show a correct first step in evaluating the expression.

$$\frac{3.2^{-4} \cdot 1}{(3.2^{-3})^2}$$

$$\frac{3.2^{-2} \cdot 3.2^0}{3.2^{-3}}$$

$$\frac{3.2^{-4}}{(3.2^{-3})^2}$$

$$\frac{3.2^{-4} \cdot 3.2^0}{3.2^{-6}}$$

- b. Evaluate $\frac{3.2^{-4} \cdot 3.2^0}{(3.2^{-3})^2}$

$$\frac{3.2^{-4} \cdot 3.2^0}{(3.2^{-3})^2}$$

$$\frac{3.2^{-4} \cdot 1}{(3.2^{-3})^2}$$

$$\frac{3.2^{-4}}{3.2^{-6}}$$

$$3.2^2$$

$$10.24$$

2. Select all the expressions that are equivalent to $\frac{1}{25}$.

☐ $(5^3) \cdot (5^{-1})^5$

☐ $\frac{(5^2)(5^5)}{(5^{-2})(5^{-4})}$

☐ $\frac{(5^0)^4 \cdot (5^{-1})}{5}$

☐ $\frac{(5^2)(5^3)}{(5^{-2})^2}$

☐ $\frac{(5^2) \cdot 5}{5^5}$



4.5.6 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with evaluating numerical expressions with exponents.



Answers will vary.

2. Explain why you chose that level of confidence.

Answers will vary.

4.5.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The data gathered in this formative assessment, coupled with observations from the lesson, provides information for differentiating instruction in subsequent lessons.

QUESTIONING

What would reflection look like for you as a teacher? Consider your own scale for how you believe students feel after the lesson and how best to address students who are not feeling very confident on evaluating numerical expressions with exponents (and/or how to address any repeated mistakes you observed being made by students during the lesson).